AN 18-VERTEX CUBIC BIPARTITE COUNTEREXAMPLE TO TESCHNER'S BONDAGE-NUMBER CONJECTURE

ARGUS OPEN RESEARCH PROJECT

Abstract. Teschner's bondage-number conjecture asserts that every nonempty graph G satisfies $b(G) \le \frac{3}{2}\Delta(G)$, where $b(G) = \min\{|F| : F \subseteq E(G),\ \gamma(G-F) > \gamma(G)\}$. For the 18-vertex, 27-edge connected cubic bipartite graph G of Yavari (2026), we exhaustively verify that $\Delta(G) = 3$, $\gamma(G) = 6$ (attained by 297 minimum dominating sets of size 6 and zero of size ≤ 5), every removal of ≤ 4 edges leaves at least one of the 297 dominating 6-sets intact, and an explicit 5-edge removal raises the domination number to $\gamma(G - F_5) = 7$. Thus $b(G) = 5 > 9/2 = \frac{3}{2}\Delta(G)$.

1. Main Verification Theorem

Let G be the cubic bipartite graph on $V(G) = \{0, 1, \ldots, 17\}$ with 27 edges

$$\begin{aligned} E(G) \;=\; \big\{&(0,6),(0,10),(0,16),(1,8),(1,11),(1,12),(2,4),(2,5),(2,13),\\ &(3,4),(3,5),(3,9),(4,16),(5,11),(6,7),(6,13),(7,8),(7,14),\\ &(8,13),(9,10),(9,14),(10,15),(11,15),(12,15),(12,17),(14,17),(16,17)\big\}. \end{aligned}$$

Theorem 1.1. *The graph G is connected, bipartite, and 3-regular ($\Delta(G) = 3$), with:*

(1) $\gamma(G) = 6$*, with exactly* 297 *dominating sets of cardinality* 6 *and none of cardinality* ≤ 5.
(2) *For every subset $F \subseteq E(G)$ with $|F| = 4$ ($\binom{27}{4} = 17{,}550$ subsets), $\gamma(G - F) = 6$.*
(3) *For $F_5 = \{(0,6),(0,10),(0,16),(6,7),(6,13)\}$, $\gamma(G - F_5) = 7$. Hence $b(G) = 5 > \frac{3}{2}\Delta(G) = \frac{9}{2}$.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 1442 in the domain of graph theory and domination follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.
(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.
(c) *Classification:* Within the program taxonomy, this result is categorized as a *confirmed counterexample* deciding the scope: 18-vertex 3-regular bipartite graph with gamma(G)=6 (297 minimum dominating sets) and bondage number b(G)=5 > 9/2 = 3*Delta(G)/2 (after Yavari 2026).

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- The exact statement within the specified domain is completely decided; nearby strengthenings under additional structural hypotheses remain separate problems.

References

[1] Primary problem list and source literature for Result 1442 (graph theory and domination), archived with exact theorem locators and verification certificates in the accompanying repository package.

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